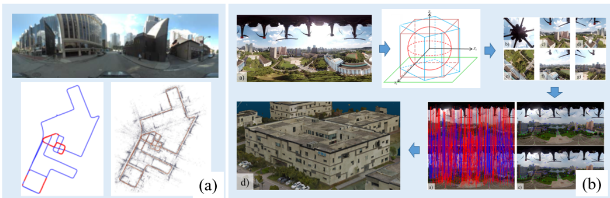

Fig. 15. SfM for spherical images. (a) direct processing of spherical images based on the unit spherical camera model [44]; (b) indirect processing of spherical images based on sphere-to-plane projection [26].

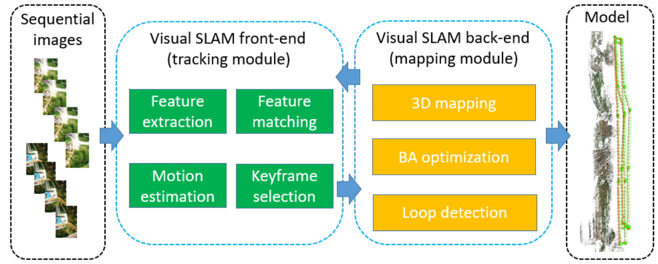

Fig. 16. The workflow of the visual SLAM system.

ature, the majority of both open-source and commercial software packages are now designed for perspective images. Table V lists the well-known and widely used software packages for 3D reconstruction in the fields of photogrammetry and computer vision. It is shown that only a few packages provide the full 3D reconstruction module for spherical images, e.g., the open-source software OpenMVG and MicMac and the commercial software Pix4Dmapper and Metashape. With the increasing popularity of spherical cameras, there is an urgent requirement for well-designed toolkits that can support scientific research and engineering application. Thus, further research is required for photogrammetric 3D reconstruction from spherical images.

3) SLAM-based online methods:

(1) Principle of visual SLAM

In contrast to the offline SfM technique, SLAM can implement simultaneous and real-time image orientation and scene reconstruction, which origins from the robotic field for localization and navigation without GNSS signals, as well as environment mapping [104]. In the field of photogrammetry and computer vision, SLAM has also been used for online image orientation, e.g., UAV and MMS images [105], [106]. Various sensors can be integrated into a SLAM system, such as RGB and depth cameras, laser scanners, GNSS and IMU (Inertial Measurement Unit) instruments, and wheel odometry [104]. For spherical image orientation, the used SLAM system is termed visual SLAM.

Fig. 16 shows the workflow of the visual SLAM, which consists of two major components. The front-end is utilized to process sequentially observed images through feature extraction, feature matching, motion estimation, and keyframe selection; the back-end is responsible for loop detection, BA optimization, and 3D mapping. The processing pipeline of the visual SLAM system includes: 1) sequentially estimating the motion of newly added images with local BA optimization; 2) checking whether or not the newly added images are keyframes and creating more map points from keyframes; 3) detecting loops between newly added images and existing 3D models and conducting global BA optimization to reduce error accumulation. Through the iterative execution of these three steps, images are sequentially oriented.

(2) SLAM with large FOV cameras

Cameras with large field-of-view angles can enhance the robustness of the SLAM systems [7] since they enable long-period feature tracking, especially in urban streets and indoor environments. Large FOV cameras have been increasingly integrated into visual SLAM systems. Recently adopted large FOV cameras can be grouped into two categories, including fisheye cameras with wide FOV and spherical cameras with full FOV.

Fisheye cameras are the most reported large FOV sensors coupled with visual SLAM systems [7], [107]–[109]. [107] proposed a direct monocular SLAM based on a unified omnidirectional camera model. It was an extension of their previous LSD-SLAM [103], and it has superior performance on the accuracy of localization and robustness to strong rotational movement. In the work of [108], the authors also designed a direct monocular SLAM system, which is an extension of the DSO system [101] and integrates the unified camera model proposed in [107]. [50] proposed a feature-based SLAM system, termed Cubemap-SLAM, based on the cubic projection of fisheye images. In this work, the cubemap model has been embedded into the ORB-SLAM. To evaluate the impact of large FOV cameras, [7] conducted a series of tests by using both indoor and outdoor datasets, and their evaluation reveals that the reconstruction accuracy also depends on the environment, and large FOV cameras tend to improve performance in indoor scenes.

Spherical cameras can further extend the FOV of fisheye cameras, and they have been widely used for data acquisition in MMS systems [24], [44]. [110] proposed a feature-based SLAM workflow that only estimates the pose of the latest image without the execution of BA optimization. The core idea is to decouple the problem of pose recovery into rotation estimation from epipolar geometry and translation estimation using 3D-2D correspondences. [102] extended the ORB-SLAM [111], [112] system into a multi-fisheye omnidirectional SLAM system, termed MultiCol-SLAM, which supports the arbitrary rigidly coupled multi-camera systems. In [6], a fisheye calibration model has been designed and used for multiple fisheye camera rigs, as shown in Fig. 17(a), and a feature-based SLAM, termed PAN-SLAM, has been proposed based on well-designed strategies for initialization, feature matching and tracking, and loop detection. The tests demonstrate the validation of the proposed SLAM system.

In recent years, spherical images captured by dual-fisheye cameras are becoming more and more popular, which promotes the development of panoramic SLAM systems [100], [113]–[115]. [100] designed a versatile visual SLAM, termed OpenVSLAM, which implements perspective, fisheye, and spherical camera models and supports monocular, stereo, and RGB-D cameras, as shown in Fig. 17(b). In addition,

TABLE V
A LIST OF OPEN-SOURCE AND COMMERCIAL SFM SOFTWARE PACKAGES.

Name	Language	Year	Website	Spherical camera
AliceVision	C++	2018	https://github.com/alicevision	×
	A photogrammetric computer vision framework for 3D reconstruction. It provides three types of camera models, i.e., Radial, Brown, and Fisheye [96].			
COLMAP	C++	2016	https://github.com/colmap/colmap	×
	A 3D reconstruction software that includes both sparse and dense matching modules, which includes Radial, and Fisheye camera models [97].			
OpenMVG	C++	2015	https://github.com/openMVG/openMVG	✓
	A well-known sparse reconstruction software that provides Radial, Brown, Fisheye, and Sphere camera models. It can directly process spherical images [21].			
MicMac	C++	2007	https://github.com/micmacIGN/micmac	✓
	An open-source photogrammetric toolkit that provides modules of AT, dense matching, and ortho-rectification for satellite, aerial and close-range images [98].			
ContextCapture	/	2022	https://www.bentley.com	×
	A well-known and widely used 3D modeling software in the field of photogrammetry, which can provide the best 3D textured mesh models. It only supports perspective cameras.			
Pix4Dmapper	/	2022	https://www.pix4d.com	✓
	A well-known and widely used photogrammetric software for aerial and close-range images in the field of photogrammetry. It supports perspective, fisheye, and spherical cameras.			
Metashape	/	2022	https://www.agisoft.com	✓
	A well-known and widely used photogrammetric software for aerial and close-range images in the field of photogrammetry. It supports perspective, fisheye, and spherical cameras.			
RealityCapture	/	2022	https://www.capturingreality.com	×
	A photogrammetric computer vision software for 3D reconstruction. It features a fast speed for image processing. It now supports a perspective camera.			

TABLE VI
A LIST OF OPEN-SOURCE SLAM SOFTWARE PACKAGES.

Name	Language	Year	Website	Spherical camera
ORB-SLAM3	C++	2021	https://github.com/UZ-SLAMLab/ORB_SLAM3	×
	A SLAM system that supports perspective and fisheye cameras and can be embedded with monocular, stereo, and RGBD cameras [99].			
OpenVSLAM	C++	2019	https://github.com/xdspacelab/openvslam	✓
	A visual SLAM system that supports various camera models, including perspective, fisheye, and spherical cameras [100].			
Cubemap-SLAM	C++	2018	https://github.com/nkwangyh/CubemapSLAM	×
	A visual SLAM system that converts fisheye images into perspective images and achieves image orientation based on the ORB-SLAM [50].			
DSO	C++	2017	https://github.com/JakobEngel/dso	×
	A direct sparse visual odometry that supports only perspective cameras [101].			
MultiCol-SLAM	C++	2016	https://github.com/urbste/MultiCol-SLAM	×
	A multi-fisheye SLAM that supports rigidly coupled multi-camera systems. It extends the ORB-SLAM and ORB-SLAM2 systems [102].			
LSD-SLAM	C++	2014	https://github.com/tum-vision/lsd_slam	×
	A direct monocular SLAM system that enables real-time image orientation and semi-dense depth map generation [103].			

the authors provided an indoor benchmark for performance evaluation [116]. [115] investigated varying feature detection and description techniques, including SPHORB and SSIFT, and compared their performance in spherical image orientation based on SLAM. Recently, [113] implemented a direct SLAM system that extends DSO (Direct Sparse Odometry) with the spherical camera model to process equirectangular images. Table VI presents a list of open-source SLAM software packages. The same finding can be made as for SfM-based methods that few SLAM systems support full spherical cameras.

E. Dense matching

Dense matching aims to establish pixel-wise correspondences and generate point clouds from SfM or SLAM-based oriented images. According to the 3D reconstruction pipeline shown in Fig. 2, generated dense point clouds would be used to construct detailed 3D models after point meshing and texture mapping. Thus, the performance of dense matching methods would determine the precision and completeness of the final 3D models. In the literature, dense matching has been an extensively studied topic with the arising of 3D reconstruction in

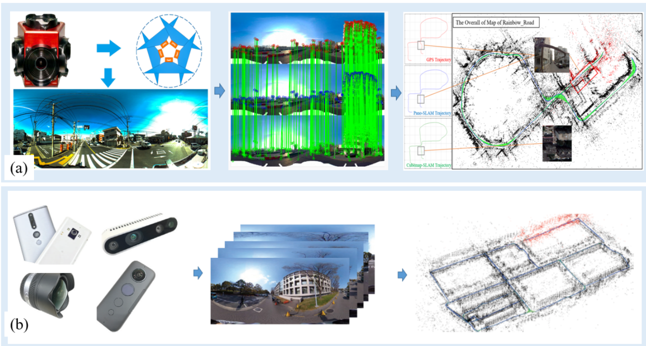

Fig. 17. The visual SLAM systems based on (a) the multi-fisheye camera rig [6] and (b) the full spherical camera [100].

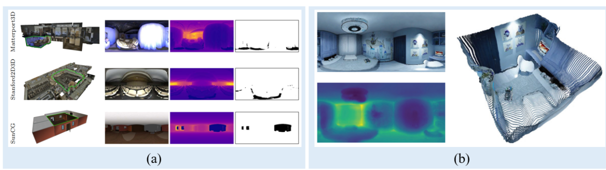

Fig. 18. Single view depth prediction for spherical images. (a) a prepared 360 dataset [28]; (b) indoor depth prediction from spherical images [119].

photogrammetry and computer vision. In addition to traditional methods for perspective images, recent years have witnessed increasingly reported dense matching methods for spherical images. Generally, this work can be grouped into single-view depth prediction and multi-view stereo matching.

1) Single-view depth prediction: Single-view depth prediction methods can process an individual spherical image instead of pixel-wise correspondence searching between images. In the literature, single-view depth prediction methods are usually implemented through learning-based CNN networks [28], [117]–[120], which belong to a new research topic in the last five years. As one of the earliest works, [28] prepared and released a spherical training dataset with ground-truth depth maps, as presented in Fig. 18(a). It can be used as the training dataset for depth estimation networks for spherical images instead of training on perspective datasets with sub-optimal performance. Considering the correlation between depth and geometric structure in indoor environments, [119] attempted to leverage existing geometric structures, e.g., corners, boundaries, and planes, as priors for network training or inferring these structures from generated depth estimation. The reconstruction models are presented in Fig. 18(b). For indoor scenes, [120] suggested representing equirectangular images by using vertical slices of the sphere, which partitions input images into vertical slices. Based on the LSTM (long short-term memory), the authors designed a network called SliceNet for depth estimation.

In contrast to indoor scenes, [117] prepared the Depth360 dataset and designed an end-to-end two-branch network, termed SegFuse, for depth prediction of spherical images. The core idea is to integrate the segmentation of cubic-map images from one branch into the depth estimation of equirectangular images from the other branch.

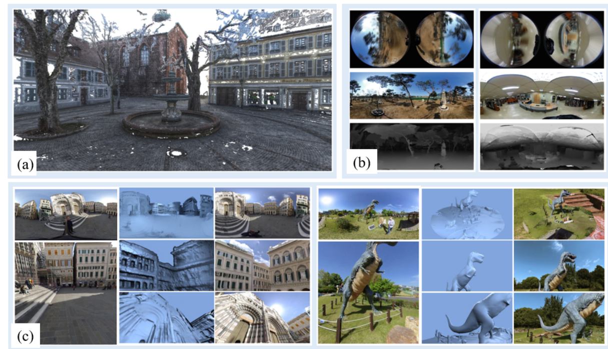

Fig. 19. multi-view dense matching for spherical images. (a) revised PMVS algorithm [121]; (b) sphere sweeping algorithm [114]; and (c) the finetuned CNN network [118].

2) Multi-view stereo matching:

(1) existing method

Multi-view stereo matching methods attempt to recover dense point clouds from two or multiple oriented images based on SfM and SLAM image orientation techniques [122]. Existing methods can be divided into two groups based on the strategies used. For the first one, traditional hand-crafted methods are on the one hand revised to adapt to spherical images. In the work of [17], [121], the PMVS (Patch-based Multi-view Stereo) library was revised to directly process spherical images for dense matching, and the results are shown in Fig. 19(a). For the second one, spherical images are projected into cubic-map images, and traditional hand-crafted algorithms and deep learning-based networks are adopted straightforward since they are designed or trained for perspective images, e.g., hand-crafted methods [123]–[126] and learning-based methods [127]–[131]. These two strategies can make use of existing well-designed methods.

(2) redesigned method

For this category, algorithms are designed and implemented from scratch considering the characteristic of spherical images [5], [29], [114], [132], [133]. In [132], a full workflow was designed for 3D reconstruction of spherical image pairs, in which a stereo matching algorithm was proposed based on a partial differential equation (PDE), and the complete 3D scene was obtained by registering partial models. Considering the power of the plane sweeping algorithm, [114] proposed a sphere sweeping algorithm based on the unified omnidirectional camera model, in which virtual spheres were used instead of virtual planes. Matching results are shown in Fig. 19(b). The algorithm was tested by using both synthetic and real datasets. These methods can be seen as mimics of the traditional dense matching algorithm for perspective images.

Recently, CNN networks have also been adopted to design multi-view stereo solutions. In the work of [29], a dual-camera imaging system that consists of top and bottom cameras was designed, which ensures that the epipolar lines of captured images are vertically aligned. A polar angle layer was added to a two-branch network for depth estimation, which plays as additional input geometric information to supervise model training. [118] proposed a complete solution for 3D reconstruction from spherical images, including image orientation,